

MACROECONOMICS

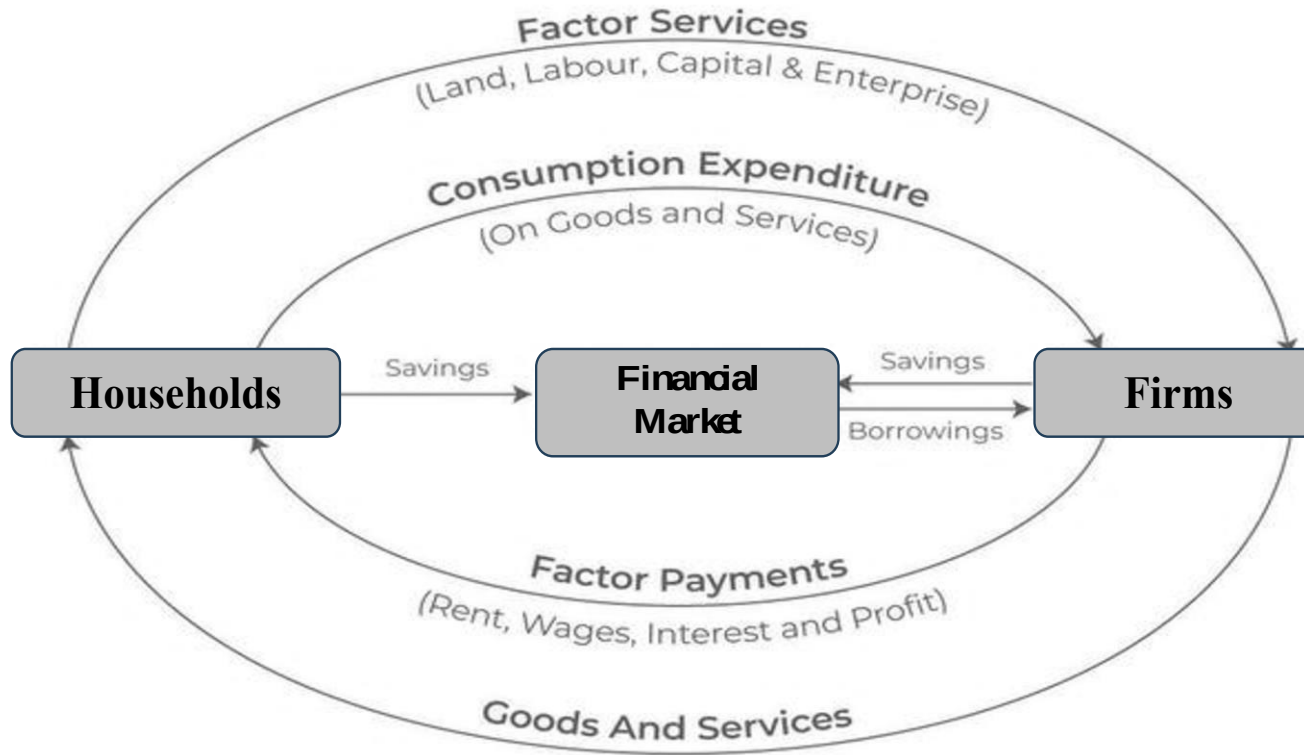
LA321/LAL200

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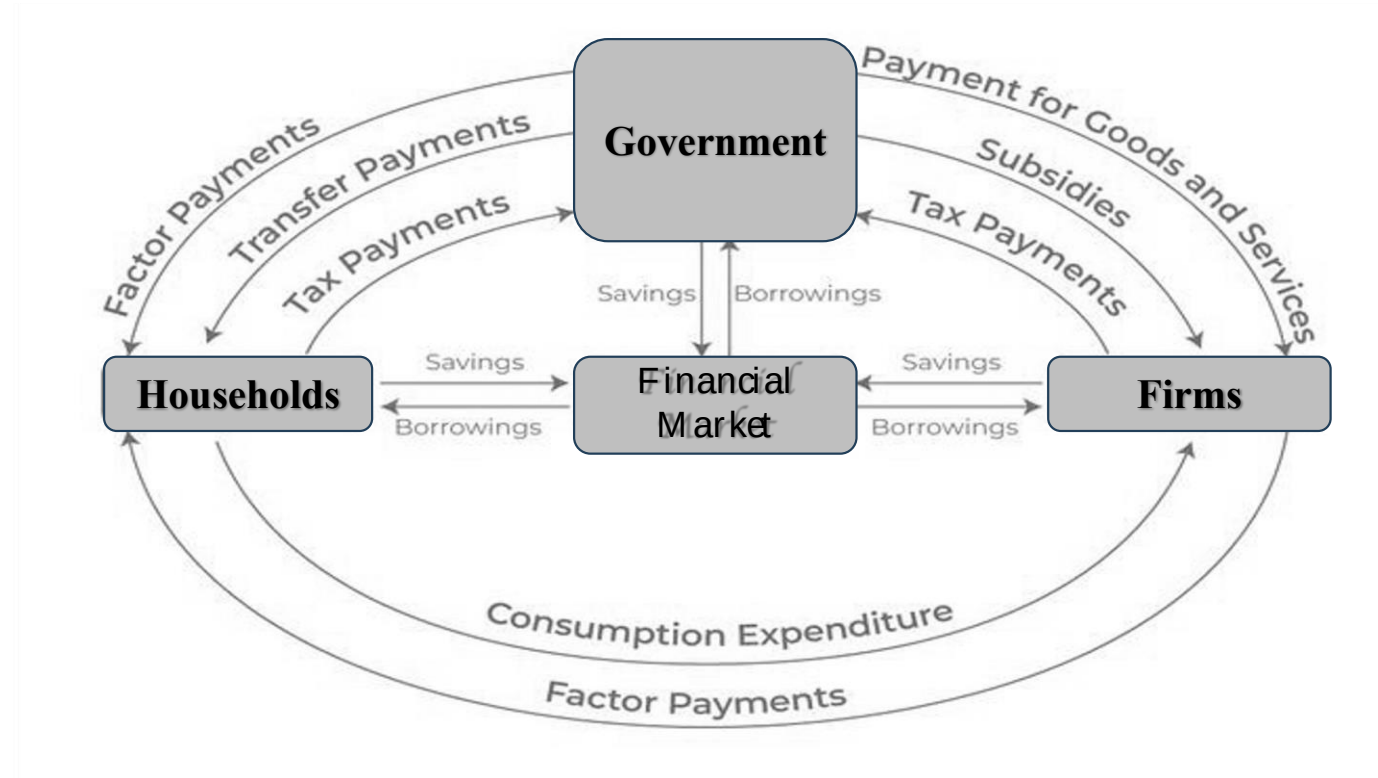
Circular Flow of Economic Activity

- **What determines the overall level of business activity? - Expenditure on firms' output.**
- **Aggregate demand** – represents the total level of spending on the goods and services produced within the country over a given time period.
- **This spending consists of four elements:**
 - Consumer spending on domestically produced goods and services (C_d),
 - Investment expenditure within the country by firms, whether on plant and equipment or on building up stocks (I),
 - Government spending on goods and services (such as health, education and transport) (G), and
 - The expenditure by residents abroad on this country's exports (X)
- **$AD = C + I + G + (X - M)$**

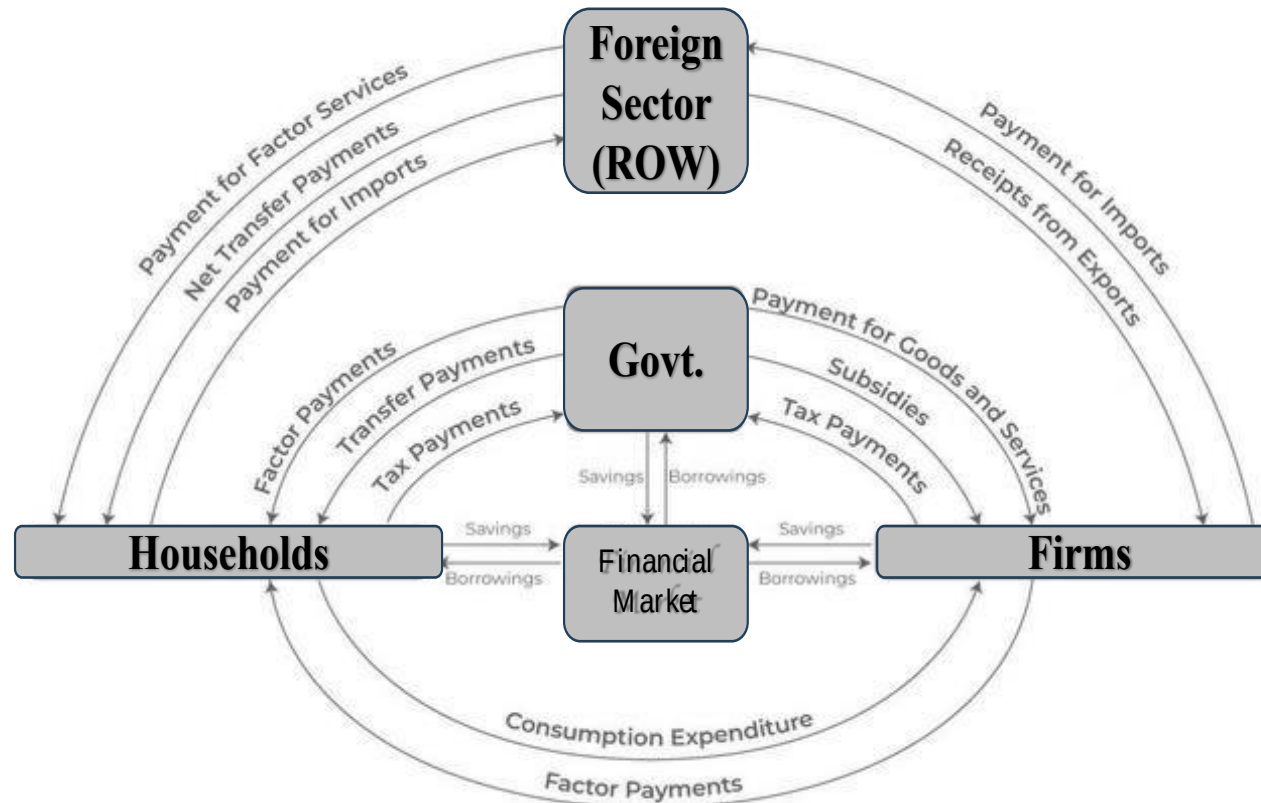
Circular Flow – Two Sector Economy



Circular Flow – Three Sector Economy



Circular Flow – Four Sector Economy



Determination of Business Activity

Identifying the equilibrium level of GDP

AGGREGATE DEMAND

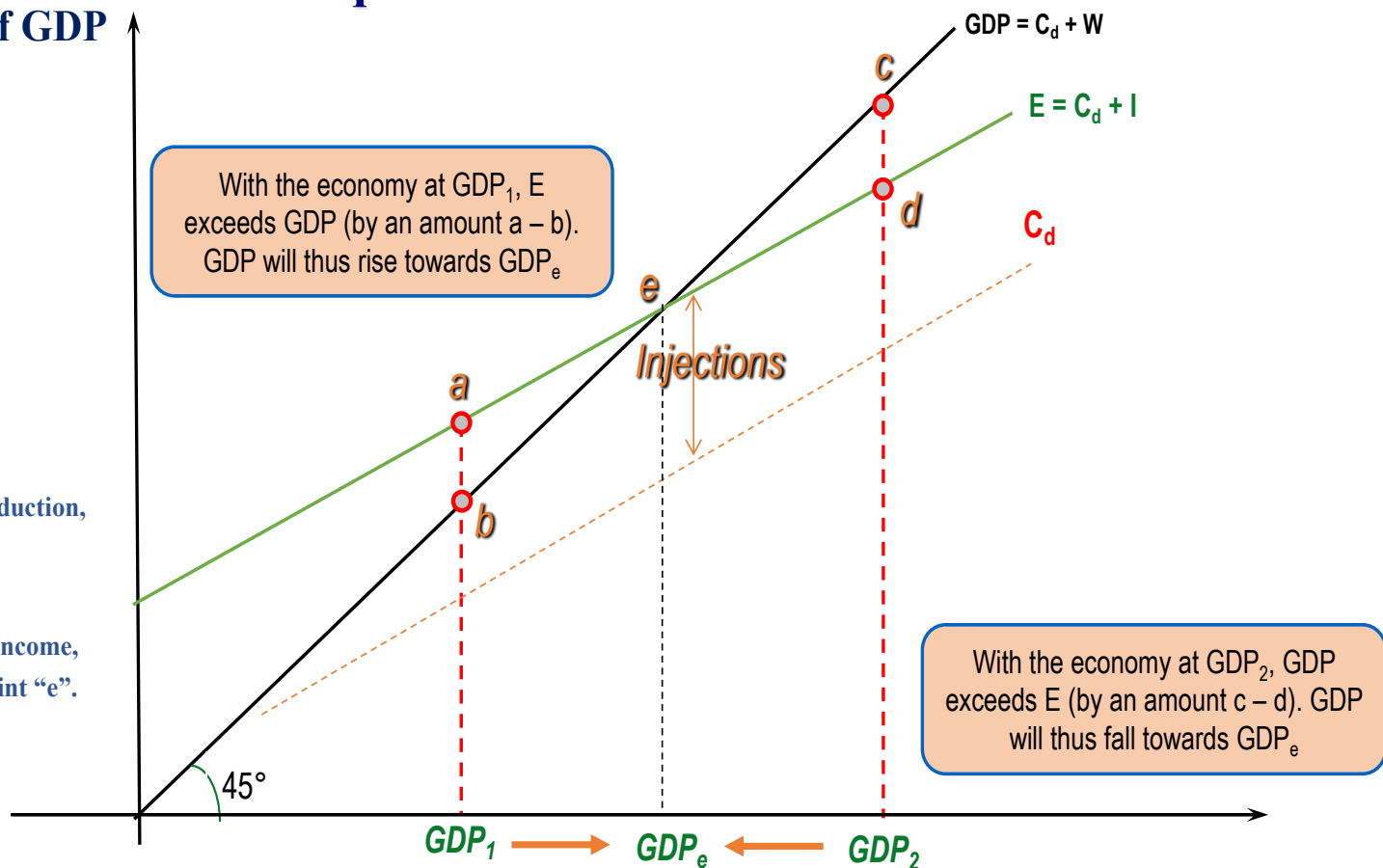
- $I > W \Rightarrow$ Rise in Exp. $\Rightarrow$ Rise in GDP
- $W > I \Rightarrow$ Fall in Exp. $\Rightarrow$ Fall in GDP

EQUILIBRIUM LEVEL

$GDP > E$: Less demand, Unsold products, Less production, Less income, Less spending and GDP falls back to “e”.

$E > GDP$: High demand, More production, Higher income, Higher spending and GDP increases to equilibrium point “e”.

Equilibrium GDP



National Income Accounting

Value Added Method	Income Method	Expenditure Method
Gross Value added from primary (GVA_p) $+$ Gross Value added from secondary sector (GVA_s) $+$ Gross Value added from tertiary sector (GVA_t) $\downarrow$ GDP_{MP} $-$ Depreciation $\downarrow$ NDP_{MP} $+$ NFIA $\downarrow$ NNP_{MP} $-$ Net Indirect Taxes $\downarrow$ NNP_{FC}	Compensation of employees $+$ Operating Surplus (Rent & royalty, Interest, Profits) $+$ Mixed Income of self-employed $\downarrow$ NDP_{FC} $+$ NFIA $\downarrow$ NNP_{FC}	Private final consumption expenditure $+$ Govt. final consumption expenditure $+$ Gross domestic capital formation $+$ Net exports $\downarrow$ GDP_{MP} $+$ NFIA $\downarrow$ GNP_{MP} $-$ Depreciation $\downarrow$ NNP_{MP} $-$ Net Indirect Taxes $\downarrow$ NNP_{FC}

Terms and Formulas

- Value Added = Value of output – intermediate consumption
- Value of output = Sales + Change in stock
- Net exports = Exports – Imports
- Gross = Net + Depreciation
- NFIA = Income earned by residents from abroad – Payments to abroad earned by foreign residents
- Factor Cost (FC) is the factor payments made to the factors of production, while MP refers to the market price at which the product is actually sold in the market.
- Net Indirect Taxes (NIT) = Indirect Taxes – Subsidies
- $GDP_{MP} - Depreciation = NDP_{MP}$
- $NDP_{MP} + NFIA = NNP_{MP}$
- $NNP_{MP} - NIT = NNP_{FC}$

Terms and Formulas

- $NNP_{FC} - NFIA = NDP_{FC}$ (**Domestic Income**)
- $NDP_{FC} - \text{Income from propert and entrepreneurship accruing to govt. administartive departments} - \text{savings of non departmental enterprises} = \text{Domestic factor income accruing to private sector}$
- Domestic factor income accruing to private sector + NFIA + Interest on national debt + Current transfers from the govt. + Net current transfers from rest of the world = **Private Income**
- Private Income – Corporate profits – Corporate taxes – Undistributed corporate profits = **Personal Income**
- $\text{Personal Income} - \text{Direct personal taxes} - \text{Misc. receipts of govt. administrative departments} = \text{Personal Disposable}$

Income

NIA

Why are the following items not included in calculating the National Income (NNP_{FC})?

- Repair and maintenance of a residential building
- Interest on car loan
- Production of services for self-consumption
- Vegetables grown in the kitchen garden
- Machine purchased for resale
- Unemployment allowance
- Prize money from game shows
- Employee's contribution to the social security schemes
- Sale of second-hand car

GDP

GDP refers to the total value of all goods and services produced within the domestic territory of the country in a year.

GDP and the growth rate hints at the future growth trajectory and economic prospects of the country. But, there are few limitations of using GDP as a welfare index :

- GDP does not account for the non-monetary exchanges
- It does not consider the negative externalities that arise as a result of growth
- Effect of inflationary situation is not considered while calculating GDP
- GDP does not account for the components of social welfare
- Though it tracks economic growth, it does not consider the income distribution pattern in the country.

Thus, we need a better index to account for the inequality that exists in the economy.

Poverty and Inequality

Measures of poverty

1. *Income-based measures*

- Poverty line: A threshold level of income adequate for meeting basic consumption needs.
 - Headcount ratio: Proportion of population living below the poverty line
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2. *Multi-dimensional measures*

- Multi-dimensional Poverty Index (MPI): Takes into account multiple dimensions of poverty such as education, health, standard of living etc.
- Human Development Index (HDI): Composite measure of a country's average score in three aspects; health, knowledge and standard of living

Poverty and Inequality

3. *Capabilities Approach*

- Developed by Amartya Sen, the approach considers individual's capabilities; their ability to do and be what they value. Absence of these capabilities indicates poverty.

4. *Absolute and Relative poverty*

- Absolute poverty shows the lack of income to meet one's own basic necessities; while relative poverty indicates the lack of income required to meet the needs and wants in comparison with others.

5. *Chronic and Transient poverty*

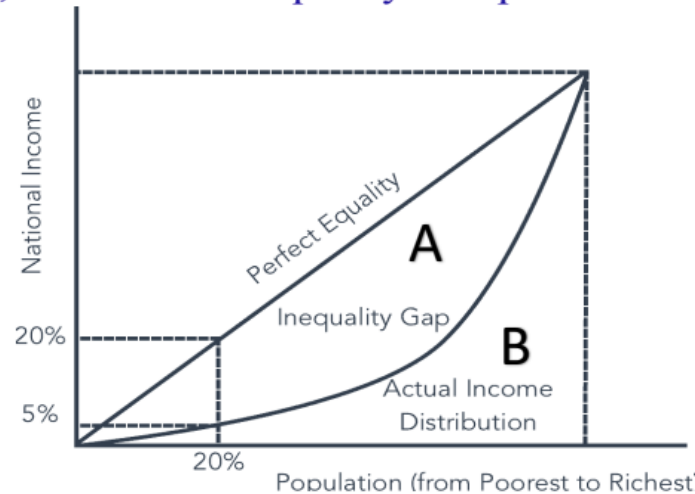
- Chronic poverty indicates long-term persistent poverty and transient poverty is the temporary poverty experienced due to some unforeseen events.

Gini Coefficient

The Gini coefficient is a measure of income distribution across a population, expressed as a number between 0 (perfect equality) and 1 (perfect inequality) .

- In general, a higher Gini index indicates greater inequality, where the higher-income strata receives the maximum share of the population's total income. A low Gini index indicates low levels of inequality.
- A **Lorenz curve** is a graphical representation of the income inequality. When the lowest 20% of the population receives only 5% of the national income, it indicates inequality as represented by the Lorenz curve.

- **Gini Coefficient** = $\frac{A}{A+B}$



GDP Deflator

Real GDP and Nominal GDP

Nominal GDP refers to the total value of goods and services produced within the domestic territory in a given year; not adjusted for inflation.

Real GDP refers to the total value of goods and services produced within the domestic territory in a given year; adjusted for inflation.

$$\text{GDP Deflator} = (\text{Nominal GDP} / \text{Real GDP}) * 100$$

$$\text{Laspeyres's Index} = \frac{\sum P_1 q_0}{\sum P_0 q_0} * 100$$

$$\text{Paasche's Index} = \frac{\sum P_1 q_1}{\sum P_0 q_1} * 100$$

$$\text{Fisher's Ideal Index} = \sqrt{\frac{\sum P_1 q_0}{\sum P_0 q_0} * \frac{\sum P_1 q_1}{\sum P_0 q_1}} * 100$$

THANK YOU